

1(a)(i)	any three from: active transport ; A active uptake against the concentration gradient ; A use of values to support using, energy / ATP ; conformational change / described (of protein carrier) ; ref. to specific binding site ;	3
1(a)(ii)	<i>similarity max 1</i> ref. to diffusion / (movement) down concentration gradient ; passive or no, energy / ATP, required ; <i>difference max 1</i> sodium ions enter by facilitated diffusion <u>and</u> oxygen molecules enter by (simple) diffusion ; A passive diffusion sodium ions, use a carrier / channel / transport, protein ; or sodium ions do not, pass through the phospholipid bilayer / cross the hydrophobic core (of the bilayer) ; A ora for oxygen	2

Question	Answer	Marks
1(a)(iii)	any two from: hydrogen carbonate ions move out of red blood cells (to be transported in plasma), so / and, chloride ions move in ; to maintain electrical neutrality (inside, cytoplasm / cell) ; A balance the charges (either side of membrane) A to prevent build-up of negative charge (in cytoplasm) AVP ; e.g. maintain equilibrium to increase reaction between carbon dioxide and water	2
1(b)	endocytosis / pinocytosis ; A bulk transport	1

Question	Answer	Marks
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